FRANKLIN (ZHENYANG) FU

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EDUCATION

University of Washington, Seattle Sept 2024 – Dec 2027 (expected)
B.S. Electrical and Computer Engineering
Relevant Coursework: Computer Architecture, Embedded Systems, Systems Programming (C/C++), Digital Circuits, Statics, Mechanics of Materials, Kinematics, Probability & Statistics
In Progress (Autumn 2026): Data-Driven Modeling and Machine Learning, Foundations of Machine Learning

PUBLICATIONS

W. Xing*, **F. Fu***, M. Han. Knowing the Answer Is Not Enough: Answer-Set Validation in Medical Question Answering. *ACM AI Letters (AILET)* — under review. (*equal contribution)

A. Wadhwani*, J. Shao*, R. Goenka*, **Z. Fu**, X. Shen, Q. Xue. Demo of NailSense: Unobtrusive Wireless Smart Nail Stickers for Whole-Hand Posture Tracking. *UIST 2026 Demo Track* — conditionally accepted. (*equal contribution)

TECHNICAL SKILLS

Programming: C, C++, Python (NumPy, Pandas), SystemVerilog
Computer Architecture & Digital Design: pipelined CPU design (forwarding, hazard detection), memory hierarchy & cache analysis, RTL simulation & verification
Embedded Systems: STM32 & ESP32 (ARM Cortex-M), FreeRTOS, sensor integration, motor control
Machine Learning & Vision: PyTorch, scikit-learn, OpenCV, YOLOv8, LLM evaluation & benchmarking
Hardware: PCB & flex-PCB prototyping, fine-pitch soldering
Mechanics & Fabrication: SolidWorks (CSWA Certified), Fusion 360, statics & stress analysis, mechanism design, 3D printing
Languages: English (professional proficiency), Chinese (native)

EXPERIENCE

UW Husky Robotics, Drone Subsystem — Embedded Firmware Developer Mar 2026 – Present
- Porting the PX4 autopilot onto an STM32H753ZI (Cortex-M7) flight controller to run flight control directly on the MCU and remove the Jetson dependency (in progress).
- Brought up a BMI270 6-axis IMU streaming calibrated data at 100 Hz, porting Bosch's SensorAPI through a custom HAL layer after tracing a board-level hardware conflict that forced an SPI-to-I2C redesign.
- Designed a dual-board architecture (STM32G431 ESC + H753 flight controller) linked over FDCAN, laying the groundwork for field-oriented motor control.

University of Washington Ubicomp Lab — Research Assistant June 2026 – Aug 2026
- Built and assembled hardware for NailSense, a flex-PCB wireless smart nail system for whole-hand posture tracking: fine-pitch soldering of nail-scale assemblies and battery charging and handling for the wireless prototypes.
- Owned the fingernail attachment mechanism across three design iterations, diagnosing each failure: silicone substrates rejected adhesives due to low surface energy; adhesive on the electronics face delaminated the flex PCB under peel force during removal; relocating the adhesive to the back surface eliminated PCB stress while improving conformity to the nail.
- Co-authored the resulting UIST 2026 demo submission (conditionally accepted).

Zhejiang University, Binjiang Institute — Research Assistant Jan 2026 – Mar 2026
- Designed Diabetes-FCT, a 1,020-item behavioral benchmark separating answer selection from answer-set validation, transforming 255 clinical seeds into paired factual-control, no-valid-option, and invalid-meta-option tasks, plus an independent 255-item fictional-premise task.
- Built an automated zero-shot evaluation pipeline in Python using regex parsing and local model deployment via Ollama, stress-testing five 7–8B models.
- Surfaced a factual-accuracy versus answer-set-validity gap: Llama-3.1-8B scored 88.2% on factual control but 23.5% when no valid option existed, while Mistral-7B fell from 81.6% to 31.0% — evidence that the two capabilities are behaviorally distinct.

UW Craft² Student Research Lab — Mechanics Leader Sept 2024 – May 2026
- Designed the hardware for HaptiNav, a haptic navigation system for blind and low-vision pedestrians: a MagSafe-compatible iPhone attachment with an ESP32-S3 BLE MCU and three directional vibration motors, running 5–7 hours on a 3.7 V Li-Po.
- Built an iOS app using ARKit LiDAR depth and camera pose reconstruction to classify obstacles (ground, drops, stairs, slopes) into an angular free-space map at up to 5 m range, transmitting avoidance commands over BLE.

- Implemented safety-first arbitration between Google Maps route guidance and the local LiDAR safety layer, overriding navigation when obstacles are detected.
- Earlier designed phone-mounted and wrist-mounted vibrotactile compass prototypes, presented at the UW Undergraduate Research Symposium.

PROJECTS

5-Stage Pipelined LEGv8 Processor — EE 469 Course Project · SystemVerilog, ModelSim Sept 2026

- Designed and verified a 5-stage pipelined LEGv8 (ARM64 subset) CPU with full forwarding and hazard detection, passing all 12 benchmarks including recursive Fibonacci.
- Built the datapath incrementally — register file, ALU, single-cycle core, then pipeline — validating each stage against predicted cycle-by-cycle waveform traces before integration.
- Reverse-engineered an undocumented cache's associativity, block size, and capacity through black-box access-latency measurement.

BinSense: Touchless Sensing Waste Bin — ESP32-S3, FreeRTOS, C++ Aug 2026

- Built a touchless waste bin on a dual-core FreeRTOS application coordinating four sensors: PIR hand detection to open a stepper-driven lid, ultrasonic fill-level sensing, photoresistor lid-state detection, and temperature/humidity monitoring to flag spoilage, with a 1602 LCD status display.
- Diagnosed a timing-corruption bug in which concurrent ultrasonic pulseIn() measurement broke DHT11 single-wire bit-bang timing; applied interrupt locking to let both tasks coexist.
- Implemented self-calibration and closed-loop actuation confirmation via photoresistor feedback; documented the codebase with Doxygen.

Automated Laser Marking System — Python, OpenCV, YOLOv8, ESP32 Feb – Mar 2026

- Architected a computer vision pipeline and ESP32-controlled laser marking gantry to localize Eppendorf tubes for medical laboratory sample-tracking automation.
- Combined YOLOv8n high-confidence anchor detection with OpenCV homography transforms to project a 13 × 15 mm 96-well grid, overcoming low-contrast lighting failures.
- Restructured custom CVAT datasets and applied Mosaic and HSV augmentation to improve dark-object recognition.
- Translated perspective-corrected physical coordinates into G-code and stepper pulse commands streamed to the ESP32 over serial.